

1) Spanning tree Protocol (STP) using Packet tracer

Aim:-

to design and Configure Spanning tree Protocol (STP) on a network with two switches connected redundantly, in order to prevent network loops and broadcast storms.

Procedure:-

1. Open Cisco packet tracer
2. Add 2 switches and 2 PCs
3. Connect the PCs to the switches
4. Connect the two switches using two ethernet cables to create a redundant path
5. Configure the PCs with suitable IP addresses
6. Open the CLI of switch 1 and configure

STP:-

```
Switch > enable
```

```
Switch # configure terminal
```

```
Switch (config) # spanning-tree
```

```
vlan 1
```

```
Switch (config) # end
```

```
Switch # show spanning-tree
```

7 Perform the same basic configuration
On switch 2

8 Use show spanning-tree to observe the
STP status

9 One redundant link will be placed into
a blocking / Alternate state to Prevent a loop

10 test connectivity between PC1 and
PC2 using Ping

Sample Output:

Switch # show spanning-tree

VLAN0001

Spanning tree enabled protocol
ieee

Root ID

Priority 32769

Address 0001.xxxx.xxxx

Interface	Role
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State

Fa0/1

Desg

Forwarding

Fa0/2

Desg

Forwarding

Fa0/3

Altn

Blocking

enul

STP was successfully configured between the two switches. The redundant path was blocked by STP, preventing network loops and broadcast storms.

2. Connecting 40 system Across two buildings

Aim:

To connect 40 system across two buildings located 200 metres apart by selecting suitable network media and topology and configuring the network using Packet tracer.

Procedure:-

1. open cisco Packet tracer
2. Divide the 40 computers into two buildings
 - Building A $\rightarrow$ 20 PCs
 - Building B $\rightarrow$ 20 PCs
3. Use switches to connect the computers within each building.
4. Use fiber-optic cable between the two building because the distance is 200 meters
5. Select star topology within each building
6. connect the switches to the PCs
7. Connect the switches of the two buildings using fiber

8 Assign IP addresses.

9 Configure the IP address of each PC.

10 Test connectivity using ping.

Sample output

PC1 > ping 192.168.2.20

Reply from 192.168.2.20: bytes = 32
time < 1ms

Reply from 192.168.2.20: bytes = 32
time < 1ms

Reply from 192.168.2.20: bytes = 32
time < 1ms

Selected media: fiber-optic cable
topology: star topology

Result -

40 systems were successfully connected across the two buildings. Star topology was used within the buildings and fiber-optic.

3 BOOTP packet analysis Using Wireshark

Aim:-

to capture and analyze BOOTP packets using Wireshark and study the header fields and payload of the concerned packets.

Procedure:

1. Open Wireshark
2. Select the active network interface
3. Start Packet capture
4. Generate BOOTP/DHCP traffic from a client.
5. Stop the capture after packets are received.
6. Apply the filter
7. Select a BOOTP Packet
8. Expand the following fields
 - Ethernet
 - IP
 - UDP
 - BOOTP
9. Analyze the source and destination IP addresses.
10. Observe the BOOTP field.

Sample output

Protocol : BOOTP / DHCP

Source port : 68

Destination port : 67

hardware type : Ethernet

transaction ID : 0xxxxxxx

client IP : 0.0.0.0

your IP : 192.168.1.10

Server IP : 192.168.1.1

Result :-

Bootp packets were successfully captured using Wireshark and their header fields and payloads information

4). Subnetting - class B Addressing

Aim :-

To illustrate subnetting using class B network address 172.16.0.0 with subnet mask 255.255.255.0 (/24).

Procedure -

1. Give network address:

172.16.0.0

2. the default class B subnet mask

255.255.0.0

or /16

3. Give subnet mask:

255.255.255.0

or /24

4. therefore, 8 bits are borrowed from the host portion

5. Number of subnets

$$2^8 = 256 \text{ subnet}$$

6. Number of usable hosts per subnet

$$2^8 - 2 = 254 \text{ host}$$

Sample output

network Address : 172.16.0.0

Subnet mask : 255.255.255.0

CIDR : /24

no. of subnets : 256

Usable host /subnet : 254

Result.

the class B network 172.16.0.0 was
Successfully subnetted using the /24
Subnet mask, Producing 256 subnets with
254 usable hosts per subnet